

Carbon Emission Tracker

Objectives

- Personalized Carbon Footprint Analysis and Visualisation.
- Predictive Analytics for Carbon Emission Reduction
- Data-Driven Recommendations

Potential objectives:

- Dynamic Carpooling Suggestions
- Gamification
 - (Through good data visualization and 'working toward goal')
 - This could include goal tracking, notifications, etc.

Feasibility Analysis for each objective

Personalized Carbon Footprint Analysis and *Visualisation*.

- Goal: Provide personalized insights into each staff member's carbon footprint.
- How? Through analysing individual and group travel patterns, and through data visualisation.

Predictive Analytics for Carbon Emission Reduction

- Goal: Predict future carbon emissions.
- How? Using a ML model -> which we will train on historical data.

Data Collection

Travel Data

Mode of Transportation

- Car, public transportation, bicycle, walking, etc.

Distance Travelled

- Kilometres or miles travelled for each journey.

Frequency of Travel

- How often the journey is made (daily, weekly, etc.).

Time of Travel

- Peak vs. off-peak hours, which could affect public transport options and car travel times due to traffic

Environmental Factors

Emission Factors

- CO2 emissions per kilometre for each mode of transportation.

Traffic Patterns (How feasible is this?)

- Data that could affect travel times and thus emissions for private vehicles.

Weather Conditions (Useful for prediction model)

- Weather can influence travel choices; for example, people might prefer public transport over biking on rainy days...

User Preferences

- Preferences for certain modes of transportation, willingness to carpool, etc.

User Interaction through the web-app

A key area which needs to be thought-out. As our solution is a web-app, it must be interactive, so the user has to be able to access and interact with it, not just view static webpages.

1. Data Input
 - Manual entry of data about their work journey today.
 - Automatically updates database.
2. Model Predictions and Suggestions
 - Have a page where ML model can process carbon estimates for the next month and can view monthly target etc.
 - This could be used to suggest alternative modes of transportation.
3. Personalised Recommendations
 - Based on the single user we could offer tailored suggestions to them.

ML Model compared with Statistical Calculations

- ML Models have potential to predict future trends by learning from historical data patterns leading to higher accuracy compared to statistical calculations.
- ML Models however need substantial amounts of quality data.
- Emphasize how using AI/ML can provide innovative solutions beyond traditional methods
- ML Models are better for forecasting future events based on historical data, because they can model complex relationships.
- For understanding and summarizing past data, statistical analysis and data visualization may be more suitable.